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APPLIED RESEARCH

Smart Office Chair for Working Conditions Optimization

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ABSTRACT There is a growing tendency for people to spend more and more hours of their day sitting. This position leads to an increase in the postural problems in the population. For workers who have to spend long hours at work, in a seated position, investment should be directed towards the development of equipment that improves their working conditions, such as smart instrumented chairs with warnings of changes in position. With this work we propose a standard office chair instrumentation approach, designed to allow monitoring physiological parameters such as: posture of the seated person (user); body temperature; and respiratory frequency. The system also allows monitoring environmental parameters such as: temperature; relative humidity; atmospheric concentration of carbon dioxide (CO₂); noise; and light levels. The chair enables the interaction with control equipment to adjust the comfort level, advising the need for rest time, repositioning notifications, and the real time visualization of data, using applications for Windows and Android. The system was tested by six users and evaluated in the detection of six different postures for each user, while sitting on the chair, presenting an 100% accuracy on the posture detection and a maximum of 18% error on the physiological parameters sensing. Experimental results show the adequate functionality of the instrumented chair, which could contribute to the prevention of pathologies associated with improper posture and the improvement of work productivity.

INDEX TERMS Ergonomic posture, pressure sensors, smart chair, temperature sensor, working conditions, work environment sensors.

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I. INTRODUCTION

Nowadays, there is a predominance of office work and with this comes an increasingly sedentary lifestyle, which has become a problem of extreme effects, as it is a leading factor in the appearance of a wide range of pathologies [1], [2].

People remain seated for excessive periods of time, both during leisure time (due to the spread of electronic entertainment, such as streaming platforms and video games) and education/work, often spending more than 15 hours per day in a seated position [3].

With the increasing tendency to stay in the sitting position for a long time, many people who suffer from back pain believe it is aggravated by their office seating conditions and sedentary lifestyle [1], [4], [5]. It has been proven that back pain is more common between the ages of 25 to 45 years old, in both genders, thus affecting people in their peak of productivity [5]. Spinal disorders are the second leading cause of requests for medical leave from work and even the reason for early retirement; consequently, they create significant losses to the economy [5], [6]. For all these reasons, many experts believe that more attention should be given to this problem, which should be studied as a social disease.

Early detection and prevention can be fundamental in mitigating such problems, which can be achieved by monitoring seated people and advising about the suitable positions to be adopted.

The seated body posture has been classified in different ways by different authors [5], [6]. There are several clinical opinions on the categorization and classification of correct and incorrect postures. Although there is not a broad basis to define it quantitatively, it can be described qualitatively based on spinal curvatures [7]. There are three possible combinations to describe an ideal sitting posture: the first portrays the thoracic and lumbar regions as flat; the second describes a long lordosis, where both regions are lordotic; and the third combination corresponds to a short lordosis that almost imitates the standing posture, with thoracic kyphosis and lumbar lordosis [7], as depicted in Figure 1. However, even considering one of these ideal positions, if it is sustained for four or more hours, it can lead to pain and discomfort for the user; it is therefore important to take short breaks to relieve and relax the muscles [1], [5], [6], [7], [8].

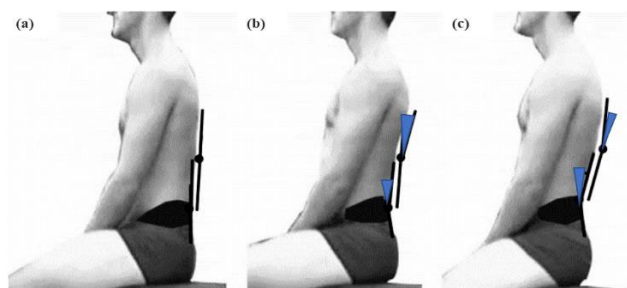


FIGURE 1. Ideal sitting postures: (a) flat, (b) long lordosis and (c) short lordosis (adapted from [7]).

Automated systems are one form of monitoring the user's position in a chair [9]. With the evolution of technology, the Internet of Things (IoT) has emerged to simplify and automate various processes [10], [11], [12]. IoT is a paradigm that aims to integrate all everyday devices through networking,

as a way of enabling autonomous communication between devices and between devices and the user [10], [11], [12]. Moreover, these devices should automatically adapt to the user's needs, improving their quality of life, which will be indispensable in the future. In addition to the IoT, artificial intelligence (AI), has also evolved to improve and provide a novel dimension to the data generated by the devices [13], [14], [15]. These two technologies are the origin of solutions which aim to monitor and warn users of misbehavior that can lead to health problems and the lack of productivity at work.

There are several systems that have been developed allowing the user's sitting posture to be monitored, using non-wearable sensors, suitable for use in the work environment [9]. In most of these projects, electronic pressure sensors are used to study posture through the pressure applied at different points of the chair. Considering the extensive list of works, only the results presented in the literature, considering pressure maps, will be highlighted here. In 2017, Huang et al. [16] reported the pressure patterns of eight standardized sitting postures of users, which were acquired by piezo-resistive sensor array and transmitted to the computer for automatic posture recognition with the application of an artificial neural network classifier. The experimental results showed that the system can recognize the eight sitting postures of humans with high accuracy.

In 2019, the implementation of a pressure map with six sensors on the seat and another map with four sensors on the backrest was reported in [17]. The data was acquired using an ESPino32 and stored in a cloud database. Another separate system reads the values stored in the database, performs pre-processing to filter the signals, and uses machine learning to recognize the user's sitting posture for five different postures. In this case, different machine learning algorithms were implemented to compare them. All the algorithms were found to have more than 90% accuracy, with the support vector machine (SVM) classifier having the highest recognition efficiency of 96.2%.

Also, in 2019 Ishaku et al. [18] reported the implementation of six flexible force sensors (two on the backrest and four on the seat) in the upholstery of an off-the-shelf office chair to enable non-intrusive monitoring of sitting postures. Besides the sensors, the system comprised an Arduino Nano microcontroller with Wi-Fi transmitter, embedded on the chair, a Wi-Fi receiver communicating with a remote server and a graphical user interface (GUI), showing real-time readings. The results show that the proposed system was able to successfully detect the nine postures tested.

In addition to pressure maps for posture recognition, other technologies using electronic components, such as pressure sensors [19], [20], accelerometers [21], video cameras [22], digital photogrammetry [23], and temperature sensors [24] have already been explored.

Incorrect postures during the time spent sitting are not the only cause of injuries, as too many hours in the same position can cause, among other, postural problems. So, there are

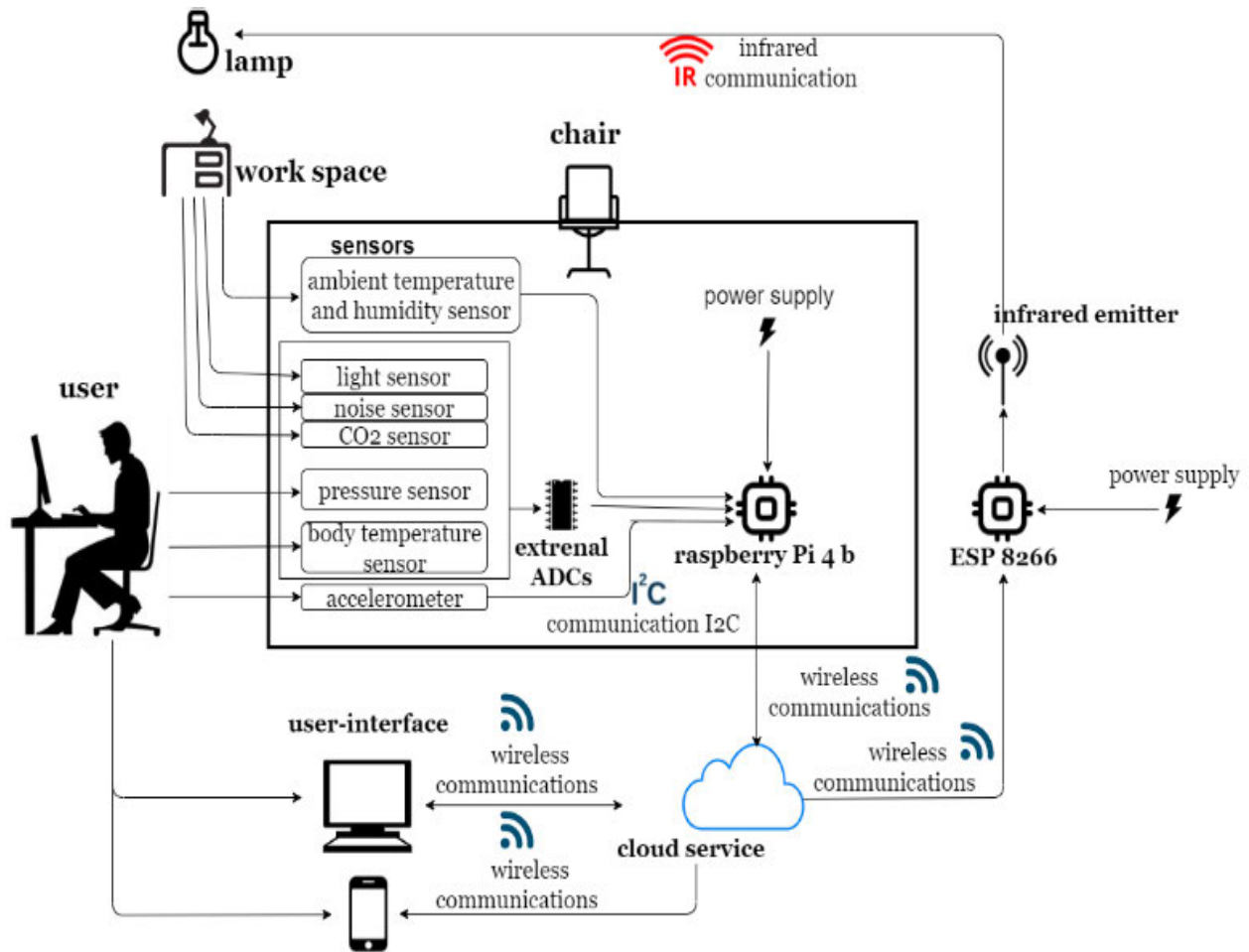


FIGURE 2. Schematic of the implemented system.

already several commercial solutions such as smartwatches that warn of moments of sedentary lifestyle, encouraging the user to take a minimum of daily steps and move with a pre-defined frequency. In addition, many of these devices are also capable of monitoring heart rate [25].

Although back pain is the main cause of discomfort for office workers, there are other factors related to the indoor environment that can also affect their performance at work, such as: excessive noise; unwanted temperatures; and insufficient light. In this sense, it is important to constantly monitor these parameters to provide a healthy and comfortable work environment, to maximize the productivity of employees at the workplace, as well as to reduce energy costs through a frequent regulation of brightness and temperature. There are some published studies on this topic that prove that both environmental conditions and the personal characteristics of workers have significant impacts on how occupants perceive their internal environment, affecting their overall satisfaction and productivity levels [26].

Our work stands out from the previous ones, as it combines in a single cost-effective solution (i.e. an instrumented chair), the monitoring and control of various parameters mentioned

above: user's posture and sitting time; physiological parameters; and parameters related to the work environment in which the user is in. Furthermore, the instrumented office chair developed in this project represents a non-intrusive and "invisible" solution [27]. Additionally, the system will be responsible for collecting, processing, and storing data in the cloud, so it can be accessed in real time by the user through his/her computer or smartphone, and consulted by the person responsible for safety in the workplace. If necessary, the user will receive alert notifications warning him/her to correct/change the posture or adjust other environmental parameters evaluated by the sensors.

The rest of the paper is organized as follows. Section II describes the general system operation. Section III focuses on sensing system calibration. The implemented system protocols, results, and discussion are presented in Section IV. Finally, the conclusion is presented in Section V.

II. GENERAL SYSTEM OPERATION

In our approach, a chair was instrumented with the aim of monitoring the following parameters: respiratory frequency; pressure to assess the user's posture, to instruct users to

change their position to avoid future injuries; body and ambient temperatures; noise level; air quality; and lighting to understand whether the workplace is providing the adequate conditions for increased productivity. The system itself can automatically control the lighting of the space to improve the comfort of the workers.

Figure 2 shows a schematic of the system implemented in the office chair, to monitor all the parameters mentioned above. In the developed system, the data is collected, processed, and sent to a database (Realtime Database) by a Raspberry Pi 4 B (RPI) microcomputer, which needs to use external analog to digital converters (ADCs). To control the room brightness, an ESP8266 microcontroller was added to the system, which is also connected to the cloud and sends commands to a lamp via an infrared transmitter, enabling the system to automatically guarantee the brightness desired/required in the room. As for the power supply, the whole system can operate using a power bank or an external power supply (220 V with 220 V AC to 5.1 V DC transformer).

The proposed overall system was tested with a power supply connected to a power outlet; however, it is possible to use a power bank battery with 5 V output, making the chair autonomous and completely free of wires. Wi-Fi connectivity was selected, as it enables the chair to communicate directly with the cloud services, independently from the existence of a base station (i.e. mobile phone or computer).

A. IMPLEMENTATION OF THE SENSORS IN THE OFFICE CHAIR

Since analog sensors require external ADCs, in our implementation, two MCP3008 ADCs were used. In total, there were 15 analog sensors implemented in the chair: four load cells on the seat for pressure monitoring; four force sensing resistors (FSR) on the backrest for pressure monitoring; one accelerometer for rate monitoring; one body temperature sensor; one humidity and temperature sensor; one noise sensor; one light sensor; and one carbon dioxide (CO₂) sensor.

The load cells were placed in the ischial tuberosity area and in the thigh area. They will be mentioned throughout the work as: SR_E (right sciatic); SL_E (left sciatic); TR_E (right thigh); and TL_E (left thigh), as schematized in Figure 3a. In this case, the “E” refers to electronic sensors.

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Four FSRs were used on the back of the chair (Figure 4a): two in the shoulder blade areas (ShR_E - right shoulder blade and ShL_E - left shoulder blade); and two others in the lumbar region (LR_E - right lumbar and LL_E - left lumbar). Although

they do not need to be calibrated, it was necessary to test several positions to find the one that provides the best response for different users. These sensors, as shown in Figure 4c, are very thin and do not provide any discomfort to the user, so they were placed directly above the sponge and under the cover seat canvas. In Figure 4 two photographs can be seen with the sensors installed in the back of the chair. Additionally, it also shows the FSR and the data acquisition system.

The sensors for physiological data monitoring, such as the user's temperature and breathing rate, were placed on the chair seat, between the sponge and the seat cover. The accelerometer was placed in the left leg area (Figure 5a). The remaining sensors for environment monitoring were attached to the back of the office chair, as shown in Figure 5b.

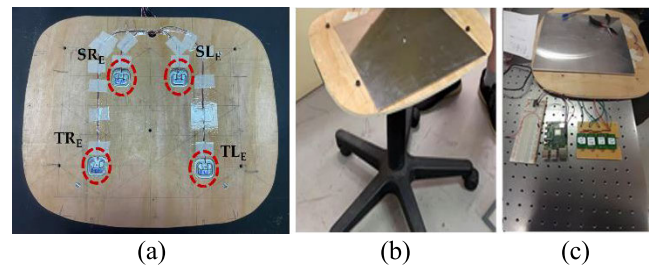


FIGURE 3. Main components of the chair seat: (a) load cell positioning; (b) aluminum sheet over wooden board for reinforcement; and (c) load cells instrumentation system.

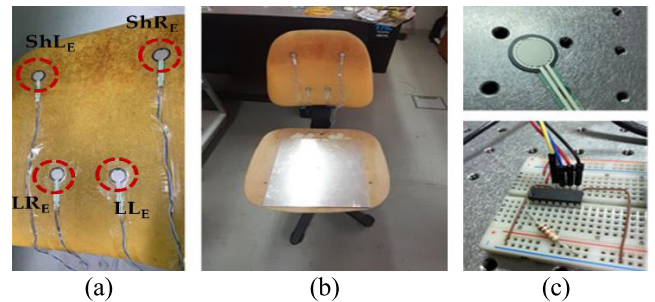


FIGURE 4. Chair backrest with FSRs: (a) FSRs positioning on the chair backrest; (b) chair with all sensors implemented (without the seat sponge); and (c) instrumentation used in the chair backrest.

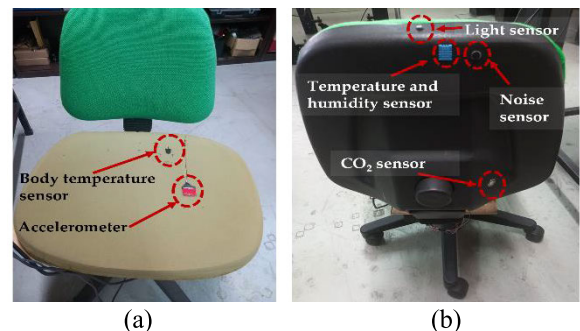


FIGURE 5. Office chair with the sensors for acquisition of physiological (a) and environmental (b) data.

B. SENSORS CHARACTERISTICS AND CALIBRATION

For most of the electronic sensors, the sensitivities used were the ones provided in their technical datasheet. For others as load cells, noise, light and CO₂ sensors, they needed to be previously calibrated.

In the case of the load cells, since they were placed into the seat of the chair, and between the load cells and the user there were various materials (like the sponge and metal plate), each load cell was placed on the chair and calibrated individually.

The range of masses used to calibrate was 0-25 kg. Since there are four sensors in the seat of the chair, a calibration of up to 25 kg allows testing the response of the sensors to an equivalent mass of 100 kg. This procedure was performed for both increasing and decreasing masses. Figure 6 shows the response of a load cell (TR_E) obtained during the calibration process. For all sensors, a linear approximation was considered. From this figure, the maximum hysteresis percentage can be calculated, which is up 8% of the full scale for the load cells.

The sensitivity coefficients obtained for all sensors are presented in Table 1. Considering the values present in this table, the load cells showed an average sensitivity of 8.02 mV/kg.

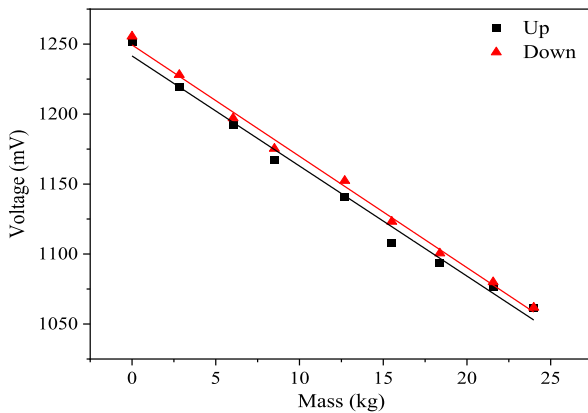


FIGURE 6. Response of the TR_E load cell to mass variation.

TABLE 1. Sensitivity coefficients.

Load cell	Sensitivity coefficients (mV/kg)
TR _E	-7.86 ± 0.31
TL _E	-11.09 ± 1.38
SR _E	-6.34 ± 0.63
SL _E	-6.79 ± 0.61

The sensors' response time was also tested. For that, a person sat down in the chair and stood up again after some time (~60 s), while the values obtained by the sensors were recorded. Figure 7 exemplifies the response of the SL_E during this test. It should be noted that the voltage difference was converted in mass values using the sensitivity coefficient presented in Table 1. To determine the response time, analysis of

the time elapsed between 10 and 90% of the sensor response to the impulse was considered. The load cell had an average response time for the rise of 470 ms and 345 ms for the fall.

For the noise and light sensors, it was necessary to make a parameterization of both data as an example to test its applicability in the environment in which we were inserted, however these parameters can be changed by the user. Thus, the analysis ranges are represented in Figure 8, and the data is read through an MCP3008 with the values in arbitrary units (AU), ranging from 0 to 65357, being 0 the ground level (0 V) and 65357 the ADC reference level (3.3 V).

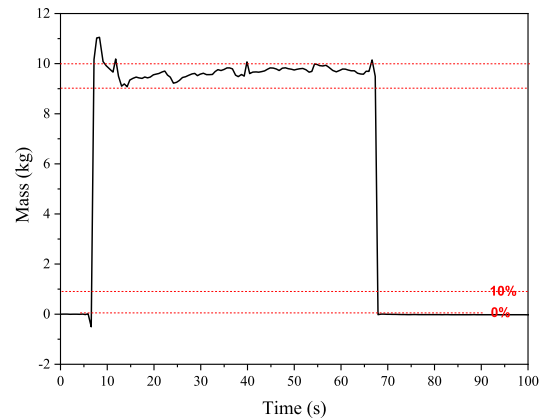


FIGURE 7. Characterization of the response time to mass variations. (sensor SL_E).

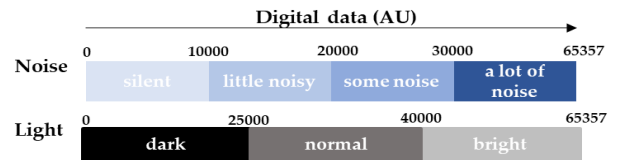


FIGURE 8. Ranges of read data for noise and light.

In the parameterization performed for demonstration purposes, in the case of the brightness values, when the system was in a room with all the windows shutters closed and ceiling lights turned on, the values obtained were standardized as the ideal working conditions related to the light. "Dark" and "Bright" were defined for values below and above the 25000 and 40000 AU range respectively, so that these two intervals were not considered ideal for a workplace.

The same logic was applied to noise. Several workplaces were tested to estimate the acceptable noise ranges. It was decided to divide the noise range into four areas. In a laboratory with working computers and other electronic machines, digital values between 13000 and 16000 AU were obtained, but it was not uncomfortable, so it was not considered disturbing and as of little concern for a workplace. With soft ambient music, values close to 26000 AU were obtained, and with very loud music the values were already above 34000 AU,

TABLE 2. Description and main characteristics of the sensors.

Sensors		Characteristics
User data	Posture	Each connected to a specially designed 24-bit HX711 ADC. The ADC circuits use a two-wire interface (clock and data) for their communication, are powered by 5 V DC, and have selectable gain. A gain of 128 was used in our implementation. Average sensitivity: 8.02 mV/kg.
	Load cells	Impedance $\sim 10 \text{ M}\Omega$ with no pressure applied to the FSRs (It decreases with an increasing force).
	FSR 402	Sampling frequency range: 1.56 to 800 Hz. Sensitivity: 500 mVg^{-1} (to dynamic acceleration ranges $\pm 2 \text{ g}$). Linear response: $10 \text{ mV}/^\circ\text{C}$; accuracy: 0.5°C (25°C). Temperature range: $-55 - 150^\circ\text{C}$.
Environmental data	Biometric data	Accelerometer – MMA8452Q Body temperature sensor – LM35
	Humidity and temperature sensor – Fenduin DHT11	Range of humidity: 20-80 % (accuracy: 5 %). Range of temperature: $0-50^\circ\text{C}$ (accuracy: 2°C).
	Noise sensor - Me Sound Sensor Makeblock v1.1	Microphone sensitivity (1 kHz): 50-54 dB.
	Light sensor – a photoresistor	Maxima values to which this sensor can be subjected: Voltage: 100 V; power: 80 mW.
	CO ₂ sensor – sen0159 CO ₂ sensor MG811	Sensor supplied with 5 V. The output voltage decreases as the CO ₂ concentration increases.

so the separation was made accordingly, as shown in Figure 8.

Regarding the CO₂ sensor, being an electrochemical sensor, it was necessary to carry out a calibration before its inclusion into the system. For this, the system was supplied with a fixed voltage of 5 V and left in continuous operation for 48 hours in a well-ventilated area. At the end of this procedure, the voltage output obtained (for CO₂ concentration values lower than 400 parts per million (ppm)) was 1.46 V. This value was later used as a reference value in the realization of the code that processes the output of the sensor.

In the case of the remaining sensors, as they were in direct contact with the environment to be monitored, no additional calibrations were required, being used the calibration data of the respective datasheets in the data treatment.

The main characteristics of the sensors are described in Table 2.

C. CONNECTIVITY AND SYSTEM SOFTWARE

The database implemented in this system is the Realtime Database (RD) provided by Google's Firebase cloud service. The software developed for this project can be divided into four sections, which include the Raspberry Pi 4 B (RPI) firmware, the ESP8266 lighting control firmware, and the computer (Windows) and smartphone (Android) applications.

Raspberry Pi 4 B

All the signal processing is performed on the RPI using Python; the processed data is then sent to the RD. For parameters that do not require a high computational load, such as temperatures, humidity and CO₂, the sensors return a voltage value, from which it is possible to determine the real value

using the sensitivity coefficient obtained beforehand for each sensor (from the calibration process). For the other sensors, the data must be processed in a more complex way so that the user only receives values that are easy to interpret.

Breathing rate is monitored using an accelerometer. Briefly, the data is filtered through an outlier removal and insert system and passes through a bandpass filter suitable for the frequencies we wish to detect. After this step, a local maximum detection method is used with a smoothed moving average filter. Further on in the Results section, the signal filtering used is described in detail.

The position adopted by the user in the chair is detected by eight sensors. The logic used for detection and data processing of these sensors starts with the conversion of the value acquired by each sensor from voltage values (V) to mass, express in Kg, using its sensitivity coefficient. Then a 50-point moving average is calculated for all sensors and then a percentage analysis of the group is performed. The analysis is accomplished separately for the seat and backrest of the chair, that is, each sensor gives a percentage value and in the case of the seat, the sum of the values measured by each sensor at each moment corresponds to 100%, which facilitates the analysis of the system. Then, according to the conditions described in Table 4, the user's posture is selected. The Raspberry Pi 4 B sends the detected posture to the RD only if it detects the same posture twice in a row. All this process takes a few seconds.

ESP8266

The ESP8266 communicates with the cloud via Wi-Fi but can also control the ambient light via infrared (IR), using the communication protocol NEC. This protocol encodes the transmitted information on a 38 kHz carrier wave in a 32-bit frame [20]. Brightness control is relatively simple;

the ESP8266 only knows two variables in the database, “Bright” and “ON/ OFF”. In the variable “ON/OFF” the ESP8266 only reads the value registered in the database and sends the command to the lighting system; if it reads “ON”, it sends the command to turn on the lights via IR and vice versa if it reads “OFF”.

In the variable “Bright”, the ESP8266 reads the recorded value and updates the value to “0” if necessary. If the value is “1” it sends the command to increase the brightness/intensity of the light system, in the case it is “-1”, a command to decrease the brightness/intensity is sent.

Computer (Windows) and Mobile (Android) application

This application enables data visualization in real time (Figure 9a). To create the interface, we used the Python Tkinter library. The interface consists of two tables and one button, as shown in Figure 9b. The top table (blue) contains information about the workplace’s environment, while the bottom table (red) contains information about the user. The button (green) refers to the brightness control, which may or may not be active.

To warn the user that he/she has been sitting incorrectly, a notification system has been implemented. This opens a pop-up window in the lower right corner of the screen with the window, as represented in Figure 9a, and plays a notification sound. This window displays the image corresponding to the frontal and profile view of the posture detected by the system. Clicking on the pop-up window opens another window with more information about the user’s posture (side and front view) and the remaining data monitored by the chair (Figure 9c).

In addition, the system has been completed with a warning related to sedentary lifestyle. If the user has been in the same position for more than 30 minutes, the device alerts him/her to take a break and to move around. This time can be adjusted according to the user’s needs (Figure 9a pop-up below).

III. OFFICE CHAIR MONITORING

A. PROTOCOL AND CHARACTERISTICS

Posture recognition tests on the office chair were performed by six adults (2 women (female – F) and 4 men (male – M)), which takes into consideration gender variation) without known pathologies and able to move without external assistance. The characteristics of the users enrolled in the test are described in Table 3.

TABLE 3. Anthropometric characteristics of the users of the chair.

Users	#1	#2	#3	#4	#5	#6
Gender	M	M	F	M	F	M
Height (cm)	172	178	167	173	165	180
Mass (kg)	77	74	59	70	54	100

The tests were performed in a closed indoor environment, at a constant temperature, in which the following sequence of

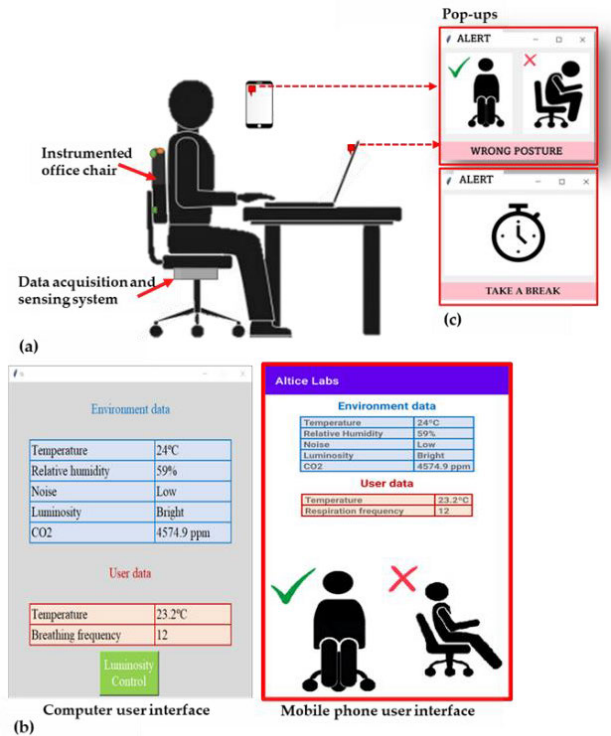


FIGURE 9. Schematic representation of the experimental setup (a), user interfaces of the computer and mobile (b), and pop-ups information (c).



FIGURE 10. Representation of the six positions adopted during the positions test.

movements was performed (Figure 10): the first position is to be correctly seated; the second to be slouching forward; the third to be slouching forward with the hips wrongly positioned; the fourth to lean back also with the hips wrongly positioned; the fifth is to be leaning on the right side; and the sixth to be leaning to the left side.

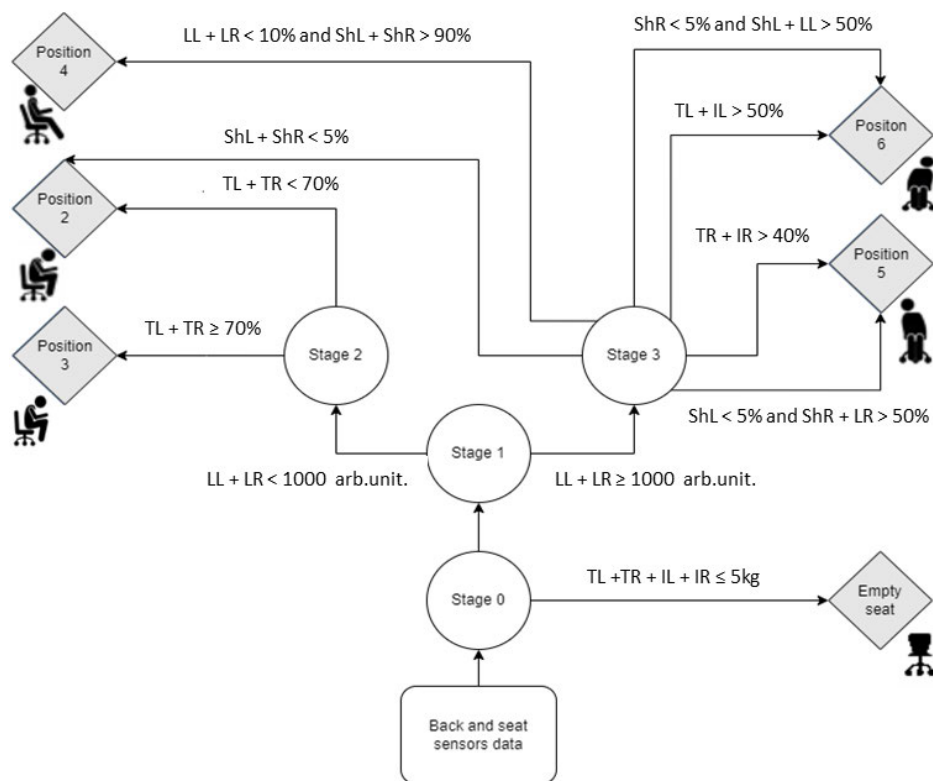
The conditions for assigning the user’s posture according to the load distribution on each sensor are described in Table 4 and the flowchart of the logic used to identify posture with the two types of sensors is shown in Figure 11.

IV. RESULTS AND DISCUSSION

Environmental and user body temperature data were experimentally validated with an external thermometer, and sensitivities are according to the manufacturers’ standards, as explained in Section II. The load distribution obtained for all the users during the test positions is presented in Table 5. To simplify the data analysis, a color code was defined that varies according to the load range in which the value is found.

TABLE 4. Conditions for assigning the user's position.

Positions	Conditions	
0	 Empty chair.	Sum seat sensors ≤ 5 kg
1	 Load is distributed among all sensors.	Sum seat sensors > 5 kg
2	 No weight is applied to the back. On the seat, the load is evenly distributed, with a slight increase in some cases in the ischial tuberosity area (I) or in the thigh area (T) depending on the inclination.	Back: $ShL + ShR + LL + LR < 5\%$ Seat: $TL + TR < 70\%$
3	 On the back, no weight is applied. On the seat, there is an increase in T area.	Back: $ShL + ShR + LL + LR < 5\%$ Seat: $TL + TR \geq 70\%$
4	 On the back, the sensors in the lumbar area (L) detect no pressure, while the sensors in the scapula area (Sh) detect almost all the pressure. On the seat, the weight distribution will tend to be more in the T zone.	Back: $LL + LR < 10\%$
5	 On the back, the ShL sensor does not detect anything. In general, the sensors on the right side will have a higher load distribution.	Back: $ShL < 5\%$ $ShR + LR > 50\%$ Seat: $TR + IR > 40\%$
6	 No load is applied in ShR sensor, in the back. In general, the sensors on the left side will have a higher load distribution.	Back: $ShR < 5\%$ $ShL + LL > 50\%$ Seat: $TL + IL > 50\%$

**FIGURE 11.** Flowchart of the logic used to identify posture with the two types of sensors.

Based on the posture detection conditions and by observing the results obtained for each user in each posture, we can

verify that the user's position is predicted with 100% confidence in the six tested positions. This means that the

TABLE 5. Percentual load distribution obtained for each of the users in each of the six positions tested and confirmation of the logic to validate the positions.

Color code by ranges of weight percentages											
		< 5 %		5 - 35 %		35 - 50 %		50 % ≤			
User 1	Position	Chair back				Chair seat				Logical Condition	Results
		ShL	ShR	LL	LR	TL	TR	IL	IR		
	1	14.8	16.6	35.8	32.8	10.0	18.8	45.0	26.2	Reference	
	2	0.0	0.0	0.0	0.0	15.8	23.2	36.4	24.6	ShL + ShR + LL + LR < 5 TL + TR < 70	✓
	3	0.0	0.0	0.0	0.0	43.1	43.5	12.7	0.7	ShL + ShR + LL + LR < 5 TL + TR ≥ 70	✓
	4	47.8	52.2	0.0	0.0	17.3	23.0	33.0	26.7	LL + LR < 10 ShL + ShR > 90	✓
	5	0.0	35.5	27.9	36.6	2.8	16.5	29.9	50.8	ShL < 5 & ShR + LR > 50 TR + IR > 40	✓
	6	27.3	0.0	45.7	27.0	7.3	8.4	70.0	14.3	ShR < 5 & ShL + LL > 50 TL + IL > 50	✓
User 2	Position	Chair back				Chair seat				Logical Condition	Results
		ShL	ShR	LL	LR	TL	TR	IL	IR		
	1	20.8	32.9	20.3	26.0	11.8	16.6	44.4	27.2	Reference	
	2	0.0	0.0	0.0	0.0	15.8	23.2	36.4	24.6	ShL + ShR + LL + LR < 5 TL + TR < 70	✓
	3	0.0	0.0	0.0	0.0	50.6	46.2	1.0	2.1	ShL + ShR + LL + LR < 5 TL + TR ≥ 70	✓
	4	46.0	54.0	0.0	0.0	29.9	37.2	17.0	16.0	LL + LR < 10 ShL + ShR > 90	✓
	5	0.0	73.8	0.0	26.2	8.6	25.1	35.7	30.7	ShL < 5 & ShR + LR > 50 TR + IR > 40	✓
	6	61.6	0.0	14.8	27.0	14.9	10.6	52.1	22.3	ShR < 5 & ShL + LL > 50 TL + IL > 50	✓
User 3	Position	Chair back				Chair seat				Logical Condition	Results
		ShL	ShR	LL	LR	TL	TR	IL	IR		
	1	15.6	15.0	35.9	33.5	20.9	27.1	33.9	18.1	Reference	
	2	0.0	0.0	0.0	0.0	24.0	25.8	31.6	18.6	ShL + ShR + LL + LR < 5 TL + TR < 70	✓
	3	0.0	0.0	0.0	0.0	49.7	45.3	3.7	1.3	ShL + ShR + LL + LR < 5 TL + TR ≥ 70	✓
	4	44.4	55.6	0.0	0.0	30.9	34.7	18.3	16.1	LL + LR < 10 ShL + ShR > 90	✓
	5	0.0	0.0	38.8	61.1	13.6	43.7	22.8	19.9	ShL < 5 & ShR + LR > 50 TR + IR > 40	✓
	6	0.0	0.0	75.5	24.5	31.0	16.4	42.3	10.3	ShR < 5 & ShL + LL > 50 TL + IL > 50	✓
User 4	Position	Chair back				Chair seat				Logical Condition	Results
		ShL	ShR	LL	LR	TL	TR	IL	IR		
	1	35.1	64.9	0.0	0.0	8.7	15.6	40.5	35.1	Reference	

TABLE 5. (Continued.) Percentual load distribution obtained for each of the users in each of the six positions tested and confirmation of the logic to validate the positions.

	2	0.0	0.0	0.0	0.0	17.1	17.4	37.6	28.0	ShL + ShR + LL + LR < 5 TL + TR < 70	✓
	3	0.0	0.0	0.0	0.0	41.6	54.6	2.7	1.1	ShL + ShR + LL + LR < 5 TL + TR ≥ 70	✓
	4	38.4	61.6	0.0	0.0	14.5	26.4	27.9	31.2	LL + LR < 10 ShL + ShR > 90	✓
	5	0.0	58.2	0.0	41.8	3.1	31.1	29.7	36.1	ShL < 5 & ShR + LR > 50 TR + IR > 40	✓
	6	27.2	0.0	44.9	27.9	19.7	6.4	60.0	13.9	ShR < 5 & ShL + LL > 50 TL + IL > 50	✓
User 5	Position	Chair back				Chair seat				Logical Condition	Results
		ShL	ShR	LL	LR	TL	TR	IL	IR		
	1	18.9	21.6	37.5	22.1	24.4	28.6	31.0	16.0	Reference	
	2	0.0	0.0	0.0	0.0	30.2	23.8	29.8	16.2	ShL + ShR + LL + LR < 5 TL + TR < 70	✓
	3	0.0	0.0	0.0	0.0	38.3	37.8	14.9	9.1	ShL + ShR + LL + LR < 5 TL + TR ≥ 70	✓
	4	49.7	50.3	0.0	0.0	14.4	18.4	41.4	25.8	LL + LR < 10 ShL + ShR > 90	✓
	5	0.0	49.2	19.7	31.1	12.5	30.3	29.6	27.6	ShL < 5 & ShR + LR > 50 TR + IR > 40	✓
	6	30.9	0.0	57.2	11.9	20.0	15.8	47.5	16.7	ShR < 5 & ShL + LL > 50 TL + IL > 50	✓
User 6	Position	Chair back				Chair seat				Logical Condition	Results
		ShL	ShR	LL	LR	TL	TR	IL	IR		
	1	16.2	27.2	28.5	28.1	8.9	7.8	51.6	31.6	Reference	
	2	0.0	0.0	0.0	0.0	9.1	7.8	44.9	38.2	ShL + ShR + LL + LR < 5 TL + TR < 70	✓
	3	0.0	0.0	0.0	0.0	48.6	48.8	1.9	0.6	ShL + ShR + LL + LR < 5 TL + TR ≥ 70	✓
	4	47.0	53.0	0.0	0.0	20.0	23.1	33.4	23.5	LL + LR < 10 ShL + ShR > 90	✓
	5	0.0	17.1	37.6	45.3	6.4	14.6	31.1	48.0	ShL < 5 & ShR + LR > 50 TR + IR > 40	✓
	6	5.1	0.0	51.3	43.6	14.2	2.8	67.6	15.4	ShR < 5 & ShL + LL > 50 TL + IL > 50	✓

positions chosen for the placement of the sensors and their selection/development work even with different heights and masses of users.

Regarding the breathing rate data, they were acquired simultaneously with the accelerometer and a BITalino (r)evolution reference device [28], [29], for testing and confirmation. After acquiring the data, it was processed following the sequence presented in the diagram of Figure 12. Initially

the raw signal passed through an outlier detector (based on a moving median system with 3-second samples and a threshold factor of 3 for removal of outliers) [30], followed by a band pass filter in the 0.1 – 0.6 Hz range. After that, the signal passed through a smoothing filter (“movemean” function in Matlab). Then, the peaks in the signal are detected using the “findpeaks” function in Matlab. One example of the results for the two types of sensors, reference sensor (BITalino)

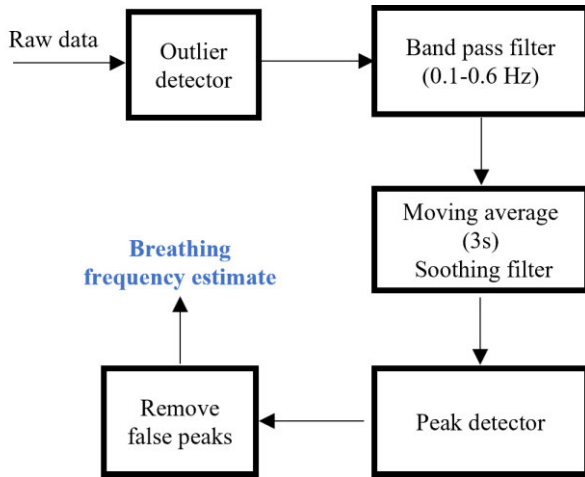


FIGURE 12. Block diagram of the breathing frequency detection algorithm.

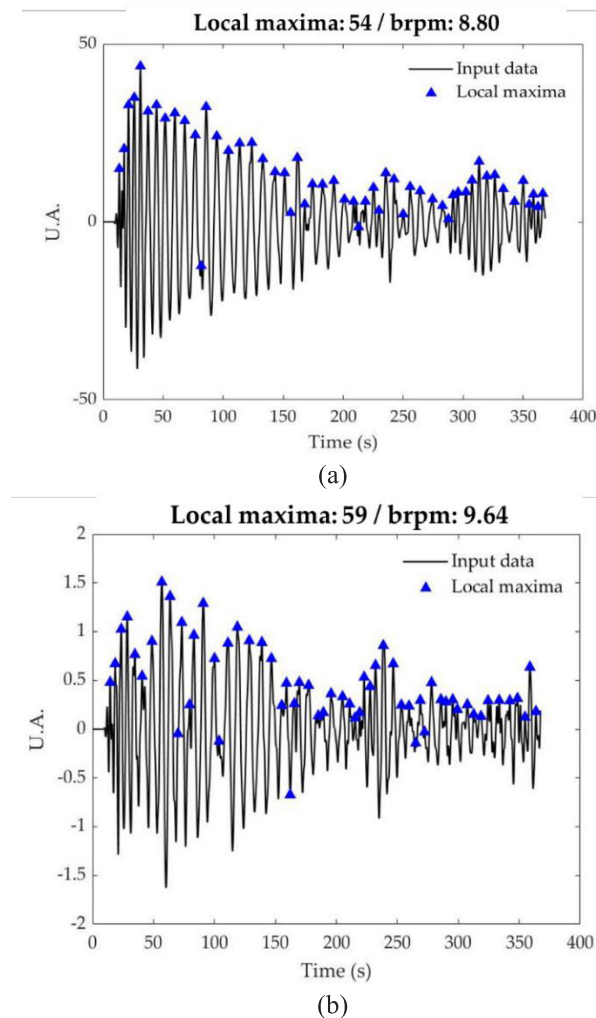


FIGURE 13. Local maxima detection to determine the breathing rate by BITalino (a) and accelerometer (b).

and accelerometer, are shown in Figure 13. Four tests were performed to analyze the breathing frequency and the results

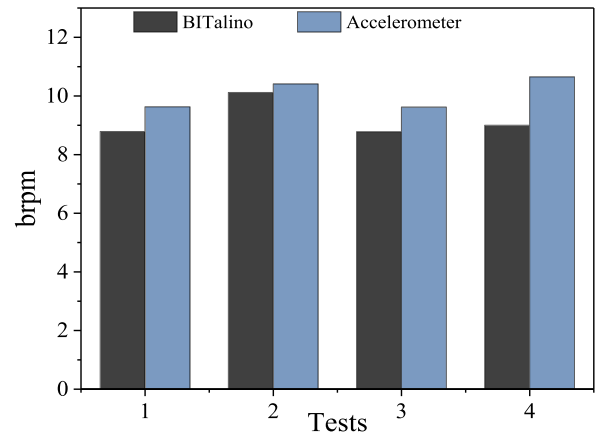


FIGURE 14. Results of breathing frequency tests for the two sensors.

for the two sensors are shown in Figure 14. All tests were performed for approximately 400 seconds. From the results presented in Figure 14, we can deduce that the two sensors have similar breathing rate values, but the accelerometer always gives slightly higher values than the reference sensor. The maximum error detected was 18%.

V. CONCLUSION

The main objective of this work was the implementation of a system in an office chair, with the ability to acquire and communicate, in electrical regime, data relating to posture, temperature and breathing frequency of the user, in addition to office environmental data, to provide better working conditions for the chair's user. Regarding the work environment, this system allows detecting environments with inappropriate noise, pollutants (CO_2) and inadequate temperatures. It warns the user through pop-ups from the accompanying app (Windows and Android) on their computer and/or mobile phone and allows automatic regulation of the space's lighting. Regarding the user's posture, this prototype was tested for a representative range of the active population, comprising volunteers with masses ranging from 54 to 100 kg, obtaining results that validate the effectiveness of the proposed solution for the detection of six different positions (five of which are improper positions), with an 100% accuracy. It is even possible to warn the user in case of exceeding a time limit in an unwanted position, or even too much time in the same position. Regarding the user's physiological data, the system permits detecting the user's body temperature and respiratory rates and warning him/her about abnormal values, with the maximum error of 18%.

It should be noted that, for accurate feedback from the smart chair, the user must define a series of initial parameters. He/she must start by calibrating the chair, placing himself in the correct position for a certain period so that the program recognizes its normal parameters. In addition, the user must define the range of light and noise that he/she considers to be correct according to the workplace where he/she is and, finally, he/she must define the frequency with which the

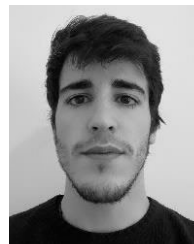
equipment will warn him/her to take a break (by default the system assumes 30 min). The user can also choose if he/she wants the pop-ups to emit sound and if he/she wants the lighting adjustment to be done automatically. It is possible to reset the parameters initially defined at any time.

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